College of Engineering and Technology
Department of Computer Engineering

Data Structures (CC215) - Data Structure & Algorithms (ECE2103)

12th week exam

Name: [redacted] Reg. Number: [redacted]

WRITE THE ANSWER IN THE LAST COLUMN

1. Suppose you are waiting for having a train ticket in a railway station. The stationmanager tells you to maintain a line for ticket. Which data structure are appropriate for maintaing a line
   A. Queue B. Stack C. Linked List D. Binary Search Tree

2. Assume we use the stack algorithm to convert the expression from infix to postfix notation. The maximum number of operators that will appear on the stack at any given time during the conversion of this infix expression: 11 + 10 * (50 * 2 - 120) is
   A. 4 B. 2 C. None of the mentioned D. 3

3. The postfix form of the expression (A+ B)*(C*D- E)*F / G is?
   A. AB + CD* E – *F *G / B. AB+ CD*E – FG /**
   C. AB + CDE * – * F *G / D. AB + CD* E – F **G /

4. Which data structure is needed to convert postfix notation to infix notation?
   A. None of the mentioned B. Tree C. All of the mentioned D. Stack

5. Which data structure is best suited for implementing recursion?
   A. Array B. stack C. Single Linked List D. Queue

6. Suppose a binary tree is constructed with n nodes, such that each node has exactly either zero or two children. The maximum height of the tree will be?
   A. (n-1)/2 B. (n+1)/2 C. (n+1)/2 -1 D. n/2 -1

7. What is the value of the postfix expression 6 3 2 4 + – *
   A. 40 B. -18 C. 74 D. 1

8. In a queue of only one element implemented using a linked list with tracked head and tail pointers, when performing the deque operation, which of these pointers will be updated?
   A. Only tail B. Both head and tail C. only heal D. None of the head or tail

```
NodeType* p;
NodeType* q;
p = new NodeType;
p->data = 12;
p->next = NULL;
q = new NodeType;
q->data = 5;
q->next = p;
```

9. Assume NodeType is a data struct representing a single linked list node. Which of the following expressions has the value NULL?
   A. p B. q→ next C. q→ next→ next D. q

10. Which of the following statement about a binary tree is CORRECT?
    A. A full binary tree is also a complete binary tree B. Any binary tree is either complete or full
    C. A complete binary tree is also a full binary tree D. A binary tree cannot be both complete and full

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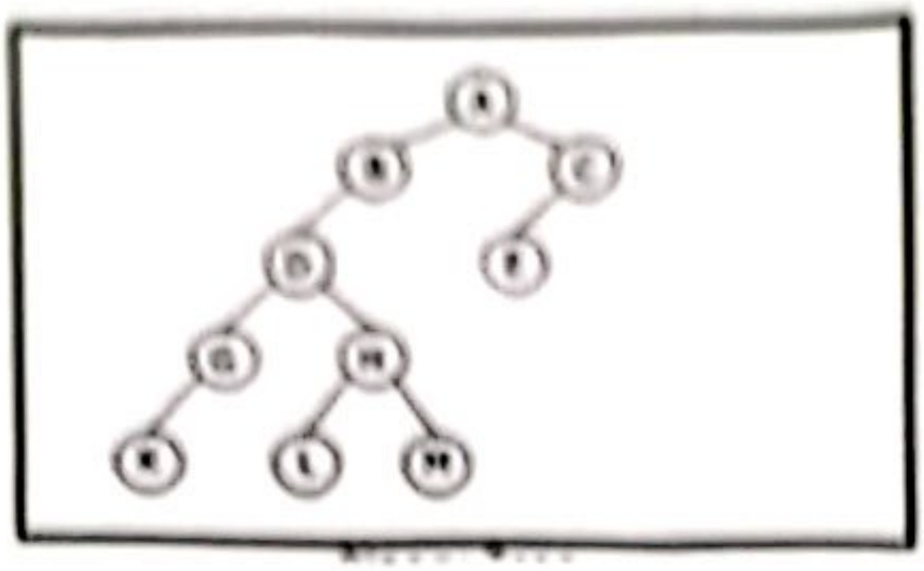

11. What is the inorder (in) traversal sequence of the above tree?
A. None of the mentioned B. K,G,D,L,H,M,B,A,E,C
C. K,G,D,H,L,M,B,A,E,C D. K,G,L,M,H,D,B,E,C,A

12. The number of nodes with degree 3,2,1,0 respectively is
A. 0,3,3,4 B. None of the mentioned C. 0,2,4,4 D. 1,2,3,4

13. The height of Root node is
A. 0 B. 2 C. 4 D. 3

14. The DFS traversal of the above tree is
A. A,B,D,G,K,H,L,M,E,C B. None of the mentioned
C. A,B,D,G,K,H,M,L,C,E D. A,B,C,D,E,G,H,K,L,M

15. The following function determines whether the input is a palindrome or not. A palindrome is a word, phrase, number, or other sequence of characters that reads the same forward and backward. Given a string, it returns 1 if palindrome and 0 otherwise, You are required to **find and correct** **<u>at least 6</u> syntax OR logical errors in the following code. For each error, circle it in the original code and re-write the line with minimal modifications to correct the error.**
Re-writing entire function, Declaring any new variables, adding new lines IS not allowed

```
struct Node { ____________________________________________________
int data; ________________________________________________________
Node* next;}; ____________________________________________________
string isPalindrome(string s) { ___________________________________
Node* stack=Null;_________________________________________________
Node* qHead =Null;________________________________________________
Node*qTail = Null_________________________________________________
for (i=0;i<len(s) ;i++) {_________________________________________
c=s[0];___________________________________________________________
push(stack, c);___________________________________________________
dequeue(qHead,qTail, c); }________________________________________
while (stack != Null || qHead != Null) {___________________________
char stackData = pop(stack);______________________________________
char queueData = dequeue(qTail,qHead,);___________________________
if (stackData == queueData) {_____________________________________
return 0;  }}_____________________________________________________
if (stack == NULL && qHead == NULL)________________________________
return 1; _________________________________________________________
```

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